

Computational Modeling of Webern's Op. 10

Standalone application - User guide

Purpose

The application is a research-informed generative score environment based primarily on Anton Webern's Five Pieces for Orchestra, op. 10. It creates a ten-staff `bach.score`, provides a built-in reference synthesis, and exports MusicXML. It is an analytical composition instrument rather than a substitute for editorial or historical source study.

Opening the application

macOS: unpack the release ZIP, move `WebernCompositionalModel.app` to a stable location, and open it. Because the research build is not notarized, the first launch may require Control-clicking the application and selecting Open.

Windows: launch `WebernCompositionalModel-Windows-x64.exe`. The single file opens a private temporary runtime, starts the program, and removes the temporary copy after the application closes. A first launch may display Microsoft SmartScreen because the research build is not code-signed; proceed only when the file's SHA-256 matches the official release note.

Max is not required on the end user's computer. No supporting files need to be placed next to the application.

Controls

Profile selects the general op. 10 synthesis or one of five differentiated movement models. **Seed** identifies a reproducible realization. Entering the same seed with the same parameters recreates the same material.

BPM defines the basic tempo. Coherence, Metamorphosis, Symmetry, Timbral contrast, Lyrical breath, Silence, Texture density, and Events govern the statistical and formal disposition of the model.

The **Activity / Time** curve controls event density across the form. The **Dynamics / Time** curve controls the global dynamic trajectory. The horizontal axis represents formal time; the vertical axis represents the respective normalized value or notated dynamic level.

Generate Material prepares a new row and internal material. **Build Score** reads the current controls and curves, then produces a complete score. The displayed row corresponds to the first aggregate realized in the score.

Play starts the internal reference synthesizer; **Stop** stops playback. **Clear** removes the current score. **Export XML** writes a MusicXML file through the operating system's save dialog.

Recommended workflow

1. Select a profile and enter a seed. 2. Set tempo and formal parameters. 3. Draw the two time curves. 4. Select Generate Material. 5. Select Build Score. 6. Inspect the score and audition it with Play. 7. Revise parameters or export MusicXML.

The built-in sound is a neutral reference synthesis. Instrumental identity, balance, articulation, and phrase direction should be evaluated from the notation and, where appropriate, in rehearsal or with dedicated orchestral playback tools.

Troubleshooting and scope

If playback is silent, stop playback, verify that the computer has an active audio output, relaunch the application, and select Play again. If MusicXML export appears incomplete, choose a new writable destination and repeat the export.

The macOS and Windows releases contain the same compositional model. Small visual differences can result from operating-system font rendering. Generated scores remain editable research outputs and should be reviewed before performance.

Technical references: Cycling '74, Standalones and Collectives, https://docs.cycling74.com/userguide/standalones_and_collectives/; bach project, <https://www.bachproject.net/>.

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